United States Patent	12419017
Kind Code	B2
Date of Patent	September 16, 2025
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Modular rugged cooled actuator control system

Abstract

A modular actuator control system comprising first and second actuator control modules configured to control first and second actuators and each having a housing, control and power electronics, power and communication connections, an actuator power connection, a flow path in each housing between an inlet and outlet port that is configured to provide a liquid coolant to the power electronics in the housing, an attachment connecting the first and second housings of the first and second actuator control modules, and a coolant inlet port of the first actuator control module configured to connect to a fluid coolant source and a coolant outlet port of the first actuator control module connected to a coolant inlet port of the second actuator control module, wherein the first actuator control module and the second actuator control module are stacked in coolant fluid communication with each other.

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Appl. No.:	**18/279426**
Filed (or PCT Filed):	**February 24, 2022**
PCT No.:	**PCT/US2022/017629**
PCT Pub. No.:	**WO2022/187066**
PCT Pub. Date:	September 09, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240196575 A1	Jun. 13, 2024

Related U.S. Application Data

us-provisional-application US 63155228 20210301

Publication Classification

Int. Cl.: **H05K7/20** (20060101)

U.S. Cl.:

CPC **H05K7/20927** (20130101); **H05K7/20272** (20130101);

Field of Classification Search

CPC: H05K (7/20927-20936); H05K (7/20272); H05K (7/20327); H05K (7/1465); H05K (7/20218); H05K (7/1471)

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to actuator control systems, and more particularly to a modular actuator control system.

BACKGROUND ART

(2) Electric motors that provide actuation in at least one motion axis are well known in the prior art

and are used in a wide variety of industries. It is known that such motors can directly or indirectly drive linear or rotary actuators or may drive a pump to provide electrohydraulic linear or rotary actuation. It is also known that such actuation systems may include drive control and power electronics to control and supervise operation of the actuator.

BRIEF SUMMARY OF THE INVENTION

(3) With parenthetical reference to the corresponding parts, portions or surfaces of the disclosed embodiment, merely for purposes of illustration and not by way of limitation, a modular actuator control system (**15**) is provided comprising: a first controller module (**18**) configured to control an electrically powered first actuator (**22**) having at least one motion axis; a second controller module (**19**) configured to control an electrically power second actuator (**23**) having at least one motion axis; the first controller module (**18**) comprising: a first housing (**30***a*); first control electronics (**32***a*) in the first housing (**30***a*); first power electronics (**31***a*) in the first housing (**30***a*); a first power connection (**36***a*) configured to connect to a power source (**67**); a first communication connection (**37***a*) configured to connect to master controller electronics (**61**); a first actuator power connection (**39***a*) configured to connect the first power electronics (**31***a*) with the first actuator (**22**); a first coolant inlet port (**80***a*, **82***a*); a first coolant outlet port (**81***a*, **83***a*); and a first flow path (**87***a*) in the first housing (**30***a*) between the first coolant inlet port (**80***a*, **82***a*) and the first coolant outlet port (**81***a*, **83***a*) that is configured to provide a liquid coolant to the first power electronics (**31***a*); the second controller module (**19**) comprising: a second housing (**30***b*); second control electronics (**32***b*) in the second housing (**30***b*); second power electronics (**31***b*) in the second housing (**30***b*); a second power connection (**36***b*) configured to connect to the power source (**67**); a second communication connection (**37***b*) configured to connect to the master controller electronics (**61**); a second actuator power connection (**39***b*) configured to connect the second power electronics (**31***b*) with the second actuator (**23**); a second coolant inlet port (**80***b*, **82***b*); a second coolant outlet port (**81***b*, **83***b*); and a second flow path (**87***b*) in the second housing (**30***b*) between the second coolant inlet port (**80***b*, **82***b*) and the second coolant outlet port (**81***b*, **83***b*) that is configured to provide the liquid coolant to the second power electronics (**31***b*); an attachment (**28***a*-**28***d*) connecting the first housing (**30***a*) of the first controller module (**18**) and the second housing (**30***b*) of the second controller module (**19**); the first coolant inlet port (**80***a*, **82***a*) of the first controller module configured to connect to a fluid coolant source (**45**); and the first coolant outlet port (**81***a*, **83***a*) of the first controller module (**18**) connected to the second coolant inlet port (**80***b*, **82***b*) of the second controller module (**19**); wherein the first controller module (**18**) and the second controller module (**19**) are stacked in coolant fluid communication with each other.

(4) The first controller module (**18**) may comprise a first secondary coolant outlet port (**81***a*) and a first secondary flow path (**85***a*) in the first housing (**30***a*) between the first coolant inlet port (**80***a*) and the first secondary coolant outlet port (**81***a*); the first flow path (**87***a*) may be separate from the first secondary flow path (**85***a*); the second controller module (**19**) may comprise a second secondary coolant inlet port (**80***b*); and the first secondary coolant outlet port (**81***a*) of the first controller module (**18**) may be connected to the second secondary coolant inlet port (**80***b*) of the second controller module (**19**); wherein the first controller module (**18**) and the second controller module (**19**) may be stacked in a parallel flow path coolant configuration (FIG. **8**).

(5) The first controller module (**18**) may comprise a first actuator communication connection (**40***a*) configured to connect the first controls electronics (**32***a*) with the first actuator (**22**); and the second controller module (**19**) may comprise a second actuator communication connection (**40***b*) configured to connect the second controls electronics (**32***b*) with the second actuator (**23**). The first actuator (**22**) may comprise a first sensor (**43***a*) for sensing an operating parameter of the first actuator (**22**) and the first actuator communication connection (**40***a*) may be configured to connect the first controls electronics (**32***a*) with the first sensor (**43***a*) of the first actuator (**22**); and the second actuator (**23**) may comprise a second sensor (**43***b*) for sensing an operating parameter of the second actuator (**23**) and the second actuator communication connection (**40***b*) may be configured

to connect the second controls electronics (**32***b*) with the second sensor (**43***b*) of the second actuator (**23**).

(6) The modular actuator control system may comprise a common power bus (**50**) supplying electric power to the first power connection (**36***a*) and the second power connection (**36***b*). The modular actuator control system may comprise a common serial bus (**60***a*) communicating with the first communication connection (**37***a*) and the second communication connection (**37***b*).

(7) The first housing (**30***a*) may comprise a first sealed electronics housing section defining a first electronics chamber (**98***a*) substantially isolated from an outside environment and the first control electronics (**32***a*) and the first power electronics (**31***a*) may be disposed in the first chamber (**98***a*); and the second housing (**30***b*) may comprise a second sealed electronics housing section defining a second electronics chamber (**98***b*) substantially isolated from the outside environment and the second control electronics (**32***b*) and the second power electronics (**31***b*) may be disposed in the second chamber (**98***b*). The first housing (**30***a*) may comprise a first connection section defining a first connection chamber (**99***a*) substantially isolated from the outside environment and the first power connection (**36***a*) and the first communication connection (**37***a*) may be disposed in the first connection chamber (**99***a*); and the second housing (**30***b*) may comprise a second connection section defining a second connection chamber (**99***b*) substantially isolated from the outside environment and the second power connection (**36***b*) and the second communication connection (**37***b*) may be disposed in the second connection chamber (**99***b*). The modular actuator control system may comprise a power bus (**50**) supplying electric power to the first power connection (**36***a*) and the second power connection (**36***b*); and a serial bus (**60***a*) communicating with the first communication connection (**37***a*) and the second communication connection (**37***b*); and the common power bus (**50**) may extend into the first connection chamber (**99***a*) and the second connection chamber (**99***b*); and the common serial bus (**60***a*) may extend into the first connection chamber (**99***a*) and the second connection chamber (**99***b*).

(8) The modular actuator control system may comprise a master controller module (**16**) comprising: a master controller housing (**30***f*); the master controller electronics (**61**) disposed in the master controller housing (**30***f*); a master communication connection (**37***f*) configured to connect to the first communication connection (**37***a*) of the first controller module (**18**) and the second communication connection (**37***b*) of the second controller module (**19**); and the attachment (**28***a*-**28***d*) connecting the first housing (**30***a*) of the first controller module (**18**), the second housing (**30***b*) of the second controller module (**19**), and the master controller housing (**30***f*) of the master controller module (**16**). The modular actuator control system may comprise a serial bus (**60***a*) communicating between the master communication connection (**37***f*), the first communication connection (**37***a*) and the second communication connection (**37***b*).

(9) The modular actuator control system may comprise a power management module (**17**) comprising: a power management housing (**30***e*); power management electronics (**51**) in the power management housing (**30***e*); an input power connection (**66**) configured to connect to the power source (**67**); an output control power connection (**37***e*) configured to connect to the first control electronics (**32***a*) and the second control electronics (**32***b*); a power coolant inlet port (**80***e*, **82***e*); a power coolant outlet port (**81***e*, **83***e*); and a power flow path (**87***e*) in the power management housing (**30***e*) between the power coolant inlet port (**80***e*, **82***e*) and the power coolant outlet port (**81***e*, **83***e*) that is configured to provide a liquid coolant to the power management electronics (**51**); and the attachment (**28***a*-**28***d*) connecting the first housing (**30***a*) of the first controller module (**18**), the second housing (**30***b*) of the second controller module (**19**), the master controller housing (**30***f*) of the master controller module (**16**), and the power management housing (**30***e*) of the power management module (**17**). The master controller module (**16**) may comprise an input control power connection (**37***f*) configured to connect to the output control power connection (**37***e*) of the power management module (**17**).

(10) The modular actuator control system may comprise a pump (**45**) connected to the first coolant

inlet port (**80***a*, **82***a*) of the first controller module (**18**) and operatively configured to pump the liquid coolant through the first flow path (**87***a*) of the first housing (**30***a*) and the second flow path (**87***b*) of the second housing (**30***b*). The modular actuator control system may comprise a heat exchanger (**46**) connected between the second coolant outlet port (**81***b*, **83***b*) of the second controller module (**19**) and the first coolant inlet port (**80***a*, **82***a*) of the first controller module (**18**). The heat exchanger (**46**) may be connected between the second coolant outlet port (**81***b*, **83***b*) of the second controller module (**19**) and the pump (**45**). The modular actuator control system may comprise a seal between the first housing (**30***a*) of the first controller module (**18**) and the second housing (**30***b*) of the second controller module (**19**).

(11) The power management module (**17**) may be stacked between the second controller module (**19**) and the master controller module (**16**) and the second coolant outlet port (**81***b*, **83***b*) of the second controller module (**19**) may be connected to the power coolant inlet port (**80***e*, **82***e*) of the power management module (**17**). The modular actuator control system may comprise: a third controller module (**20**, **21**) configured to control an electrically power third actuator (**24**, **25**, **26**, **27**) having at least one motion axis; the third controller module (**20**, **21**) comprising: a third housing (**30***c*, **30***d*); third control electronics (**32***c*, **32***d*) in the third housing (**30***c*, **30***d*); third power electronics (**31***c*, **31***d*, **31***e*, **31***f*) in the third housing (**30***c*, **30***d*); a third power connection (**36***c*, **36***d*) configured to connect to the power source (**67**); a third communication connection (**37***c*, **37***d*) configured to connect to the master controller electronics (**61**); a third actuator power connection (**39***c*, **39***d*, **39***e*, **39***f*) configured to connect the third power electronics (**31***c*, **31***d*, **31***e*, **31***f*) with the third actuator (**24**, **25**, **26**, **27**); a third coolant inlet port (**80***c*, **80***d*, **82***c*, **82***d*); a third coolant outlet port (**81***c*, **81***d*, **83***c*, **83***d*); and a third flow path (**87***c*, **87***d*) in the third housing (**30***c*, **30***d*) between the third coolant inlet port (**80***c*, **80***d*, **82***c*, **82***d*) and the third coolant outlet port (**81***c*, **81***d*, **83***c*, **83***d*) that is configured to provide the liquid coolant to the third power electronics (**31***c*, **31***d*, **31***e*, **31***f*); the attachment (**28***a*-**28***d*) connecting the first housing (**30***a*) of the first controller module (**18**), the second housing (**30***b*) of the second controller module (**19**), and the third housing (**30***c*, **30***d*) of the third controller module (**20**, **21**); and the second coolant outlet port (**81***b*, **83***b*) of the second controller module (**19**) connected to the third coolant inlet port (**80***c*, **80***d*, **82***c*, **82***d*) of the third controller module (**20**, **21**); wherein the first controller module (**18**), the second controller module (**19**), and the third controller module (**20**, **21**) are stacked in coolant fluid communication with each other. The second controller module (**19**) may comprise a second secondary coolant outlet port (**83***b*) and a second secondary flow path (**86***b*) in the second housing (**30***b*) between the second coolant inlet port (**82***b*) and the second secondary coolant outlet port (**83***b*); the second flow path (**87***b*) may be separate from the second secondary flow path (**86***b*); the third controller module (**21**) may comprise a third secondary coolant inlet port (**82***d*); and the second secondary coolant outlet port (**83***b*) of the second controller module (**19**) may be connected to the third secondary coolant inlet port (**82***d*) of the third controller module (**21**); wherein the first controller module (**18**) and the second controller module (**19**) may be stacked in a series flow path coolant configuration and the second controller module (**19**) and the third controller module (**21**) may be stacked in a parallel flow path coolant configuration.

(12) The first flow path (**87***a*, **287**) in the first housing (**30***a*) between the first coolant inlet port (**82***a*, **282**) and the first coolant outlet port (**81***a*) may comprise a plurality of separate coolant passages (**90**, **290**) in thermal proximity to the first power electronics (**31***a*). The first flow path (**187**) in the first housing (**30***a*) between the first coolant inlet port (**180**) and the first coolant outlet port may comprise a serpentine coolant passage in thermal proximity to the first power electronics (**31***a*).

(13) The master controller housing (**30***f*) of the master controller module (**16**) may contain a braking resistor (**53**) and a DC capacitor (**54**). The power source (**67**) may comprise an electric-vehicle battery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a front isometric view of an embodiment of an improved modular actuator control system.
(2) FIG. **2** is a rear isometric view of the modular actuator control system shown in FIG. **1**.
(3) FIG. **3** is a schematic view of the modular actuator control system shown in FIG. **1** operationally coupled to an example liquid cooling system and multiple example actuators.
(4) FIG. **4** is a top plan schematic view of the modular actuator control system shown in FIG. **1**.
(5) FIG. **5** is a left side isometric view of the left actuator control module shown in FIG. **1**.
(6) FIG. **6** is a partial vertical lateral sectional view of the actuator control module shown in FIG. **5** illustrating a first embodiment internal cooling conduit profile.
(7) FIG. **7** is a vertical longitudinal sectional schematic view of the actuator control system shown in FIG. **1** in a series cooling stacked configuration.
(8) FIG. **8** is a vertical longitudinal sectional schematic view of the actuator control system shown in FIG. **1** in a parallel cooling stacked configuration.
(9) FIG. **9** is a vertical longitudinal sectional schematic view of the actuator control system shown in FIG. **1** in a combined serial and parallel cooling stacked configuration.
(10) FIG. **10** is a left side isometric view of the power management module shown in FIG. **1**.
(11) FIG. **11** is an alternative second embodiment internal cooling conduit profile to the first embodiment shown in FIG. **6**.
(12) FIG. **12** is an alternative second embodiment internal cooling conduit profile to the first embodiment shown in FIG. **6**
DETAILED DESCRIPTION OF THE EMBODIMENTS
(13) At the outset, it should be clearly understood that like reference numerals are intended to identify the same structural elements, portions or surfaces consistently throughout the several drawing figures, as such elements, portions or surfaces may be further described or explained by the entire written specification, of which this detailed description is an integral part. Unless otherwise indicated, the drawings are intended to be read (e.g., crosshatching, arrangement of parts, proportion, degree, etc.) together with the specification, and are to be considered a portion of the entire written description of this invention. As used in the following description, the terms “horizontal”, “vertical”, “left”, “right”, “up” and “down”, as well as adjectival and adverbial derivatives thereof (e.g., “horizontally”, “rightwardly”, “upwardly”, etc.), simply refer to the orientation of the illustrated structure as the particular drawing figure faces the reader. Similarly, the terms “inwardly” and “outwardly” generally refer to the orientation of a surface relative to its axis of elongation, or axis of rotation, as appropriate.
(14) Referring now to the drawings, and more particularly to FIGS. **1-4**, a modular actuator control system is provided, a first embodiment of which is generally indicated at **15**. As shown, system **15** comprises central control module **16**, central power module **17**, first actuator control module **18**, second actuator control module **19**, third actuator control module **20**, and fourth actuator control module **21**, which are all stacked together to provide compact, rugged, and cooled controller system **15** for controlling a plurality of actuators **22-27**.
(15) As shown, first actuator control module **18** controls linear electromechanical actuator **22**, second actuator control module **19** controls rotary electrohydraulic actuator **23**, third actuator control module **20**, which is a dual drive controller, controls both rotary electromechanical actuator **24** and rotary electromechanical actuator **25**, and fourth actuator control module **21**, which is a dual drive controller, controls both linear electrohydraulic actuator **26** and rotary electromechanical actuator **27**. While this embodiment includes four actuator control modules **18-21** that control six actuators **22-27**, other configurations may be employed depending on the desired application. For

example, and without limitation, less than four or more than four actuator control modules may be stacked together as desired. In addition, and without limitation, the actuator control modules may be configured to control alternative types of actuators. Thus, the modular system is easily adaptable and expandable and may comprise different stacked actuator control modules depending on the desired actuator count and functionality.

(16) As shown in FIGS. **1-4**, in this embodiment actuator control module **18** is a single axis controller and generally comprises housing **30***a* containing power electronics **31***a*, controller electronics **32***a* and interior coolant conduit **87***a* in a front portion, and containing actuator power connection **36***a* operatively connected to DC bus **50** and communication connection **37***a* operatively connected to communications and control power bus **60***a* and **60***b* in a rear portion.

(17) As shown, housing **30***a* is a sturdy enclosure generally comprising front panel **100***a*, left side panel **102***a* with back-end opening cover **106***a*, right side panel **103***a* with back end opening **106***b*, bottom panel **105***a*, top panel **104***a*, and rear panel **101***a* with rear opening cover **107***a*. As mentioned above, housing **30***a* generally comprises front compartment **98***a*, which generally houses control electronics **32***a*, power electronics **31***a* and interior cooling conduit **87***a*, and rear compartment **99***a*, which generally houses the connectivity to the other modules. The front and rear compartments **98** and **99** are separated by an interior panel with properly sized openings for power and communication bus connections. The housing panels are punctured by various connections and by various opening that may be plugged when not used. As shown in FIG. **1**, front panel **100***a* of housing **30***a* includes actuator power connections **39***a*, actuator sensor feedback connections **40***a*, and in this embodiment auxiliary interface **41***a*. Actuator power connection **39***a* provides driving power to actuator **22**. Actuator sensor feedback connection **40***a* interfaces with the feedback sensors **43***a* of actuator **22**. Auxiliary interface **41***a* allows for the connection of module **18** to other external auxiliary sensors and functions if desired. Housing **30***a* of module **18** protects the internal electronics from the outside environment when stacked with modules **16**, **17** and **19-21** as shown. While in this embodiment rear panel **101***a* is shown with a rear access opening that is covered by rear opening cover **107***a* after assembly of the modular stack, alternatively and without limitation rear panel **101***a* may be a solid panel.

(18) As shown, power electronics **31***a* provide operational power to the terminals of electric motor **22***a* via actuator power connection **39***a*. Power electronics **31***a* converts DC power from connection **36***a* to DC bus **50** into a controlled Pulse Width Modulated (PWM) current which drives motor **22***a*. The operation of power electronics **31***a* is governed by PWM control signals from power control interface **35***a* of motor control electronics **32***a*.

(19) Motor control electronics **32***a* control, monitor and supervise operation of actuator **22**, including the control of power to motor **22***a* via power control interface **35***a*. Motor control electronics **32***a* include communication interface **33***a*, processor **34***a*, and power control interface **35***a*. Communication interface **33***a* provides communications with central control module **16** and if desired other actuator control modules **19**, **20** and **21** via communication connection **37***a*. Communication interface **33***a* communicates data, commands and states. Processor **34***a* provides internal control and monitoring. Processor **34***a* receives commands from central controller module **16** and feedback from sensors **43***a* recording operating parameters of actuator **22**, via actuator sensor connection **40***a*, and controls actuator **22** accordingly. In this embodiment, such sensors are coupled to control electronics **32***a* via wired connection **40***a*. In other embodiments, they may be coupled via a wireless connection. Processor **34***a* is configured to perform a variety of computer-implemented functions such as performing method steps, calculations and the like and storing relevant data. Processor **34***a* may be any digital device which has output lines that are a logic function of its input lines, examples of which include a microprocessor, microcontroller, field programmable gate array (FPGA), programmable logic device (PLD), application specific integrated circuit, or other similar devices.

(20) In this embodiment, actuator **22** is a linear electromechanical actuator having three-phase

permanent magnet DC electric motor **22***a* driving output shaft **22***b*. Linear magnetic motor **22***a* includes a stationary stator and a sliding shaft that is driven to move linearly (that is, as a straight line translation) with respect to the stator. The shaft is at least partially surrounded by the stator and is held in place relative to the stator by a bearing. The shaft generates a magnetic field by virtue of having a series of built in permanent magnets. The stator generates magnetic fields through annular magnetic coils. By timing the flow of current in the coils with respect to the position and/or momentum of the shaft, the interaction of magnetic forces from the shaft and from the stator will actuate the shaft to move linearly in either direction. Other motors may also be used as alternatives. A position sensor **43***a* provides position feedback, via connection **40***a*, to monitor shaft position which is used for closed loop motion control in motor control electronics **32***a*. A position sensor may be any electrical device for measuring the position, or a derivative of position, or distance from an object, examples of which include an encoder, a resolver, a linear variable differential transformer, a variable resistor, a variable capacitor, a laser rangefinder, an ultrasonic range detector, an infrared range detector, or other similar devices.

(21) As shown in FIGS. **1-4**, in this embodiment actuator control module **19** is a single axis controller and, similar to module **18**, generally comprises housing **30***b* containing power electronics **31***b*, controller electronics **32***b*, and interior coolant conduit **87***b* in a front portion and containing actuator power connection **36***b* operatively connected to DC bus **50** and communication connection **37***b* operatively connected to communication and control power bus **60** in a rear portion.

(22) As shown, housing **30***b* is a sturdy enclosure generally comprising front panel **100***b*, left side panel **102***b* with back-end opening **106***c* aligned with back end opening **106***b* in right side panel **103***a* of module **18**, right side panel **103***b* with back end opening **106***d*, bottom panel **105***b*, top panel **104***b*, and rear panel **101***b* with rear opening cover **107***b*. As mentioned above, housing **30***b* generally comprises front compartment **98***b*, which generally houses control electronics **32***b*, power electronics **31***b* and interior cooling conduit **87***b*, and rear compartment **99***b*, which generally houses the connectivity to the other modules. The front and rear compartments **98***b* and **99***b* are separated by an interior panel with properly sized openings for power and communication bus connections. The housing panels are punctured by various connections and by various opening that may be plugged when not used. As shown in FIG. **1**, front panel **100***b* of housing **30***b* includes actuator power connections **39***b*, actuator sensor feedback connections **40***b*, and in this embodiment auxiliary interface **41***b*. Actuator power connection **39***b* provides driving power to actuator **23**. Actuator sensor feedback connection **40***b* interfaces with the feedback sensors **43***b* of actuator **23**. Auxiliary interface **41***b* allows for the connection of module **19** to other external auxiliary sensors and functions if desired. Housing **30***b* of module **19** protects the internal electronics from the outside environment when stacked with modules **16**, **17**, **18** and, **20** and **21** as shown. While in this embodiment rear panel **101***b* is shown with a rear access opening that is covered by rear opening cover **107***b* after assembly of the modular stack, alternatively and without limitation rear panel **101***b* may be a solid panel.

(23) As shown, power electronics **31***b* provide operational power to the terminals of electric motor **23***a* via actuator power connection **39***b*. Power electronics **31***b* converts DC power from connection **36***b* to DC bus **50** into a controlled PWM current which drives motor **23***a*. The operation of power electronics **31***b* is governed by PWM control signals from power control interface **35***b* of motor control electronics **32***b*.

(24) Motor control electronics **32***b* control, monitor and supervise operation of actuator **23**, including the control of power to motor **23***a* via power control interface **35***b*. Motor control electronics **32***b* include communication interface **33***b*, processor **34***b*, and power control interface **35***b*. Communication interface **33***b* provides communications with central control module **16** and if desired other actuator control modules **18**, **20** and **21** via communication connection **37***b*. Communication interface **33***b* communicates data, commands and states. Processor **34***b* provides internal control and monitoring. Processor **34***b* receives commands from central controller module

16 and feedback from sensors **43***b* recording operating parameters of actuator **23**, via actuator sensor connection **40***b*, and controls actuator **23** accordingly. In this embodiment, such sensors are coupled to control electronics **32***a* via wired connection **40***a*.

(25) In this embodiment, actuator **23** is a rotary electrohydraulic actuator generally comprising variable speed bidirectional electric servomotor **23***a* and bi-directional or reversible pump **23***b* driven by motor **23***a*. In this embodiment motor **23***a* is a brushless D.C. variable-speed servo-motor that is supplied with a current. Motor **23***a* has an inner rotor with permanent magnets and a fixed non-rotating stator with coil windings. When current is appropriately applied through the coils of the stator via power electronics **31***b* and power connection **39***b*, a magnetic field is induced. The magnetic field interaction between the stator and the rotor generates torque which may rotate the output shaft of motor **23***a*. Control electronics **32***b*, based on position feedback via connection **40***b*, generate and commutate the stator fields via power electronics **31***b* to vary the speed and direction of motor **23***a*. Accordingly, motor **23***a* will selectively apply a torque on its output shaft in either direction about the output shaft axis at varying speeds. Other motors may be used as alternatives. For example, a variable speed stepper motor, brush motor or induction motor may be used. In this embodiment pump **23***b* is a fixed displacement bi-directional internal two-port gear pump. The pumping elements, namely intermeshed gears, are capable of rotating in either direction, thereby allowing hydraulic fluid to flow in either direction. This allows for oil to be added into and out of the system as controller **32***b* closes the control loop of position or pressure. At least one gear of pump **23***b* is connected to the output shaft of motor **23***a*, with the other pump gear following. The direction of flow of pump **23***b* depends on the direction of rotation of the rotor and the output shaft of motor **23***a*. The speed and output of pump **23***b* is variable with variations in the speed of motor **23***a*. Other bi-directional pumps may be used as alternatives. For example, a variable displacement pump may be used.

(26) As shown in FIGS. **1-4**, in this embodiment actuator control module **20** is a dual axis controller and generally comprises housing **30***c* containing power electronics **31***c*, power electronics **31***d*, controller electronics **32***c*, and interior coolant conduit **87***c* in a front portion and containing actuator power connection **36***c* operatively connected to DC bus **50** and communication connection **37***c* operatively connected to communication and control power bus **60** in a rear portion.

(27) As shown, housing **30***c* is a sturdy enclosure generally comprising front panel **100***c*, left side panel **102***c* with back-end opening **106***e* aligned with back end opening **106***d* in right side panel **103***b* of module **19**, right side panel **103***c* with back end opening **106***f*, bottom panel **105***c*, top panel **104***c*, and rear panel **101***c* with rear opening cover **107***c*. As mentioned above, housing **30***c* generally comprises front compartment **98***c*, which generally houses control electronics **32***c*, power electronics **31***c*, power electronics **31***d*, and interior cooling conduit **87***c*, and rear compartment **99***c*, which generally houses the connectivity to the other modules. The front and rear compartments **98***c* and **99***c* are separated by an interior panel with properly sized openings for power and communication bus connections. The housing panels are punctured by various connections and by various opening that may be plugged when not used. As shown in FIG. **1**, front panel **100***c* of housing **30***c* includes first actuator power connection **39***c*, second actuator power connection **39***d*, first actuator sensor feedback connection **40***c*, second actuator sensor feedback connection **40***d*, and in this embodiment auxiliary interface **41***c*. Actuator power connection **39***c* provides driving power to actuator **24**. Actuator sensor feedback connection **40***c* interfaces with the feedback sensors **43***c* of actuator **24**. Actuator power connection **39***d* provides driving power to actuator **25**. Actuator sensor feedback connection **40***d* interfaces with the feedback sensors **43***d* of actuator **25**. Auxiliary interface **41***c* allows for the connection of module **20** to other external auxiliary sensors and functions if desired. Housing **30***c* of module **20** protects the internal electronics from the outside environment when stacked with modules **16**, **17**, **18**, **19** and **21** as shown. While in this embodiment rear panel **101***c* is shown with a rear access opening that is covered by rear opening cover **107***c*

after assembly of the modular stack, alternatively and without limitation rear panel **100***c* may be a solid panel.

(28) As shown, power electronics **31***c* provide operational power to the terminals of electric motor **24***a* via actuator power connection **39***c*. Power electronics **31***c* converts DC power from connection **36***c* to DC bus **50** into a controlled PWM current which drives motor **24***a*. The operation of power electronics **31***c* is governed by PWM control signals from power control interface **35***c* of motor control electronics **32***c*. As shown, power electronics **31***d* provide operational power to the terminals of electric motor **25***a* via actuator power connection **39***d*. Power electronics **31***d* converts DC power from connection **36***c* to DC bus **50** into a controlled PWM current which drives motor **25***a*. The operation of power electronics **31***d* is governed by PWM control signals from power control interface **35***c* of motor control electronics **32***c*.

(29) Motor control electronics **32***c* control, monitor and supervise operation of actuator **24** and actuator **25**, including the control of power to motor **24***a* via power control interface **35***c* and control of power to motor **25***a* via power control interface **35***c*. Motor control electronics **32***c* include communication interface **33***c*, processor **34***c*, and power control interface **35***c*. Communication interface **33***c* provides communications with central control module **16** and if desired other actuator control modules **18**, **19**, and **21** via communication connection **37***c*. Communication interface **33***c* communicates data, commands and states. Processor **34***c* provides internal control and monitoring. Processor **34***c* receives commands from central controller module **16** and feedback from sensors **43***c* recording operating parameters of actuator **24**, via actuator sensor connection **40***c*, and controls actuator **24** accordingly. Processor **34***c* also receives commands from central controller module **16** and feedback from sensors **43***d* recording operating parameters of actuator **25**, via actuator sensor connection **40***d*, and controls actuator **25** accordingly.

(30) In this embodiment, actuators **24** and **25** are rotary electromechanical actuators generally comprising variable speed bidirectional electric servomotors **24***a* and **25***a*, respectively. In this embodiment motors **24***a* and **25***a* are brushless D.C. variable-speed servo-motors with electronically controlled commutation systems that are supplied with a current and include resolver feedback to monitor rotor angle which is used for closed loop motion control in actuator control electronics **32***c*. Motor **24***a* has an inner rotor with permanent magnets and a fixed non-rotating stator with coil windings. When current is appropriately applied through the coils of the stator via power electronics **31***c* and power connection **39***c*, a magnetic field is induced. The magnetic field interaction between the stator and the rotor generates torque which may rotate the output shaft of motor **24***a*. Control electronics **32***c*, based on position feedback via connection **40***c*, generate and commutate the stator fields via power electronics **31***c* to vary the speed and direction of motor **24***a*. Similarly, motor **25***a* has an inner rotor with permanent magnets and a fixed non-rotating stator with coil windings. When current is appropriately applied through the coils of the stator via power electronics **31***d* and power connection **39***d*, a magnetic field is induced. The magnetic field interaction between the stator and the rotor generates torque which may rotate the output shaft of motor **25***a*. Control electronics **32***c*, based on position feedback via connection **40***d*, generate and commutate the stator fields via power electronics **31***d* to vary the speed and direction of motor **25***a*. Accordingly, motors **24***a* and **25***a* will each selectively apply a torque on its output shaft in either direction about the output shaft axis at varying speeds. Other motors may be used as alternatives. For example, a variable speed stepper motor, brush motor or induction motor may be used.

(31) As shown in FIGS. **1-4**, in this embodiment actuator control module **21** is a dual axis controller and, similar to module **20**, general comprises housing **30***d* containing power electronics **31***e*, power electronics **31***f*, controller electronics **32***d*, and interior coolant conduit **87***d* in a front portion and containing actuator power connection **36***d* operatively connected to DC bus **50** and communication connection **37***d* operatively connected to communication and control power bus **60** in a rear portion.

(32) As shown, housing **30***d* is a sturdy enclosure generally comprising front panel **100***d*, left side

panel **102***d* with back-end opening **106***g* aligned with back end opening **106***f* in right side panel **103***c* of module **20**, right side panel **103***d* with back end opening **106***h*, bottom panel **105***d*, top panel **104***d*, and rear panel **101***d* with rear opening cover **107***d*. As mentioned above, housing **30***d* generally comprises front compartment **98***d*, which generally houses control electronics **32***d*, power electronics **31***e*, power electronics **31***f*, and interior cooling conduit **87***d*, and rear compartment **99***d*, which generally houses the connectivity to the other modules. The front and rear compartments **98***d* and **99***d* are separated by an interior panel with properly sized openings for power and communication bus connections. The housing panels are punctured by various connections and by various opening that may be plugged when not used. As shown in FIG. **1**, front panel **100***d* of housing **30***d* includes first actuator power connection **39***e*, second actuator power connection **39***f*, first actuator sensor feedback connection **40***e*, second actuator sensor feedback connection **40***f*, and in this embodiment auxiliary interface **41***d*. Actuator power connection **39***e* provides driving power to actuator **26**. Actuator sensor feedback connection **40***e* interfaces with the feedback sensors **43***e* of actuator **26**. Actuator power connection **39***f* provides driving power to actuator **27**. Actuator sensor feedback connection **40***f* interfaces with the feedback sensors **43***f* of actuator **27**. Auxiliary interface **41***d* allows for the connection of module **21** to other external auxiliary sensors and functions if desired. Housing **30***d* of module **21** protects the internal electronics from the outside environment when stacked with modules **16**, **17**, **18**, **19** and **20** as shown. While in this embodiment rear panel **101***d* is shown with a rear access opening that is covered by rear opening cover **107***d* after assembly of the modular stack, alternatively and without limitation rear panel **101***d* may be a solid panel.

(33) As shown, power electronics **31***e* provide operational power to the terminals of electric motor **26***a* via actuator power connection **39***e*. Power electronics **31***e* converts DC power from connection **36***d* to DC bus **50** into a controlled PWM current which drives motor **26***a*. The operation of power electronics **31***e* is governed by PWM control signals from power control interface **35***d* of motor control electronics **32***d*. As shown, power electronics **31***f* provide operational power to the terminals of electric motor **27***a* via actuator power connection **39***f*. Power electronics **31***f* converts DC power from connection **36***d* to DC bus **50** into a controlled PWM current which drives motor **27***a*. The operation of power electronics **31***f* is governed by PWM control signals from power control interface **35***d* of motor control electronics **32***d*.

(34) Motor control electronics **32***d* control, monitor and supervise operation of actuator **26** and actuator **27**, including the control of power to motor **26***a* via power control interface **35***d* and control of power to motor **27***a* via power control interface **35***d*. Motor control electronics **32***d* include communication interface **33***d*, processor **34***d*, and power control interface **35***d*. Communication interface **33***d* provides communications with central control module **16** and if desired other actuator control modules **18**, **19**, and **20** via communication connection **37***d*. Communication interface **33***d* communicates data, commands and states. Processor **34***d* provides internal control and monitoring. Processor **34***d* receives commands from central controller module **16** and feedback from sensors **43***e* recording operating parameters of actuator **26**, via actuator sensor connection **40***e*, and controls actuator **26** accordingly. Processor **34***d* also receives commands from central controller module **16** and feedback from sensors **43***f* recording operating parameters of actuator **27**, via actuator sensor connection **40***f*, and controls actuator **27** accordingly.

(35) In this embodiment, actuator **26** is a linear electrohydraulic actuator having electric motor **26***a* driving hydraulic pump **26***b* in a closed loop hydraulic circuit to extend and retract hydraulic cylinder driving mechanism **26***c*. In this embodiment, servo-motor **26***a* is used to drive reversible pump **26***b* to extend and retract piston **26***d* in cylinder **26***c*, with pump **26***b* pressurizing a working fluid, typically hydraulic oil, directly raising the pressure in a hydraulic gap on one side or the other of hydraulic piston **26***d*. In this embodiment motor **26***a* is a brushless D.C. variable-speed servo-motor that is supplied with a current. Motor **26***a* has an inner rotor with permanent magnets and a fixed non-rotating stator with coil windings. When current is appropriately applied through the

coils of the stator via power electronics **31***e* and power connection **39***e*, a magnetic field is induced. The magnetic field interaction between the stator and the rotor generates torque which may rotate the output shaft of motor **26***a*. Control electronics **32***d*, based on position feedback via connection **40***e*, generate and commutate the stator fields via power electronics **31***e* to vary the speed and direction of motor **26***a*. Accordingly, motor **26***a* will selectively apply a torque on its output shaft in either direction about the output shaft axis at varying speeds. In this embodiment pump **26***b* is a fixed displacement bi-directional internal two-port gear pump. The pumping elements, namely intermeshed gears, are capable of rotating in either direction, thereby allowing hydraulic fluid to flow in either direction. This allows for oil to be added into and out of the system as controller **32***d* closes the control loop of position or pressure. At least one gear of pump **26***b* is connected to the output shaft of motor **26***a*, with the other pump gear following. The direction of flow of pump **26***b* depends on the direction of rotation of the rotor and the output shaft of motor **26***a*. The speed and output of pump **26***b* is variable with variations in the speed of motor **26***a*.

(36) In this embodiment, actuator **27** is a rotary electromechanical actuator generally comprising variable speed bidirectional electric servomotor **27***a*. In this embodiment motor **27***a* is a brushless D.C. variable-speed servo-motors with an electronically controlled commutation system that is supplied with a current and includes resolver feedback to monitor rotor angle which is used for closed loop motion control in actuator control electronics **32***d*. Motor **27***a* has an inner rotor with permanent magnets and a fixed non-rotating stator with coil windings. When current is appropriately applied through the coils of the stator via power electronics **31***f* and power connection **39***f*, a magnetic field is induced. The magnetic field interaction between the stator and the rotor generates torque which may rotate the output shaft of motor **27***a*. Control electronics **32***d*, based on position feedback via connection **40***f*, generate and commutate the stator fields via power electronics **31***f* to vary the speed and direction of motor **27***a*. Accordingly, motor **27***a* will selectively apply a torque on its output shaft in either direction about the output shaft axis at varying speeds.

(37) As shown in FIGS. **1-4**, in this embodiment central power management module **17** generally comprises housing **30***e* containing control power electronics **55**, auxiliary battery electronics **52**, and interior coolant conduit **87***e* in a front portion, and containing actuator supply power connection **36***e* operatively connected to DC bus **50** and control power connection **37***e* operatively connected to control power bus **60***b* in a rear portion.

(38) As shown, housing **30***e* is a sturdy enclosure generally comprising front panel **100***e*, left side panel **102***e* with back-end opening **106***i* aligned with back end opening **106***h* in right side panel **103***d* of module **21**, right side panel **103***e* with back end opening **106***j*, bottom panel **105***e*, top panel **104***e*, and rear panel **101***e* with rear opening cover **107***e*. As mentioned above, housing **30***e* generally comprises front compartment **98***e*, which generally houses control power electronics **55**, auxiliary battery electronics **52**, and interior coolant conduit **87***e*, and rear compartment **99***e*, which generally houses the connectivity to the other modules. The front and rear compartments **98***e* and **99***e* are separated by interior panel **58** with properly sized openings for power and control bus connections. The housing panels are punctured by various connections and by various opening that may be plugged when not used. As shown in FIG. **1**, front panel **100***e* of housing **30***e* includes DC bus power input connection **66**, auxiliary battery connection **71**, and in this embodiment auxiliary low power connection **72**. DC bus power input connection **66** is configured to connect to power source **67**, which in this embodiment comprises an electric vehicle main battery pack. Auxiliary battery connection **71** is configured to connect to auxiliary battery **70**, which in this embodiment comprises a lower voltage battery, such as for example a 12 volt battery. Low power connection **72** allows for the connection of module **17** to other external low voltage devices if desired. Housing **30***e* of module **17** protects the internal electronics from the outside environment when stacked with modules **16**, and **18-21** as shown. While in this embodiment rear panel **101***e* is shown with a rear access opening that is covered by rear opening cover **107***e* after assembly of the modular stack,

alternatively and without limitation rear panel **101***e* may be a solid panel.

(39) As shown, control power electronics **55** provide lower voltage operational power, via connection **37***e* and control power bus **60***b*, to actuator control electronics **32***a*, **32***b*, **32***c* and **32***d* of modules **18**, **19**, **20** and **21**, and master control electronics **61** of module **16**. Auxiliary battery electronics **52** are connected to external auxiliary battery **70** via auxiliary battery connection **71** to power the system logic prior to main power source **67** being active.

(40) As shown in FIGS. **1-4**, in this embodiment central control module **16** general comprises housing **30***f* containing master control electronics **61**, regenerative braking electronics **53** and DC capacitor **54** in a front portion and containing power connection **36***f* operatively connected to DC bus **50** and communication connection **37***f* operatively connected to communication and control power bus **60***a* and **60***b* in a rear portion.

(41) As shown, housing **30***f* is a sturdy enclosure generally comprising front panel **100***f*, left side panel **102***f* with back-end opening **106***k* aligned with back end opening **106***j* in right side panel **103***e* of module **17**, right side panel **103***f*, bottom panel **105***f*, top panel **104***f*, and rear panel **101***f* with rear opening cover **107***f*. As mentioned above, housing **30***f* generally comprises front compartment **98***f*, which generally houses master control electronics **61**, regenerative braking electronics **53** and DC capacitor **54**, and rear compartment **99***f*, which generally houses the connectivity to the other modules. The front and rear compartments **98***f* and **99***f* are separated by an interior panel with properly sized openings for power and communication bus connections. The housing panels are punctured by various connections and by various opening that may be plugged when not used. As shown in FIG. **1**, front panel **100***f* of housing **30***f* includes external hardwired communication bus connection **64**, external wireless antenna interface **65**, USB port **68**, ethernet port **69**, and in this embodiment auxiliary interfaces **41***f*. Housing **30***f* of module **16** protects the internal electronics from the outside environment when stacked with modules **17-21** as shown. While in this embodiment rear panel **101***f* is shown with a rear access opening that is covered by rear opening cover **107***f* after assembly of the modular stack, alternatively and without limitation rear panel **101***f* may be a solid panel.

(42) DC capacitor **54** smooths out power on common bus **50** from electronic high frequency switching and braking resistor **53** dissipates regenerative power from actuators **22-27**. Master control electronics **361** control, monitor and supervise operation of individual actuator control modules **18-21**. Central control electronics **61** include internal communication interface **62**, master processor **63**, external wired communication interface **64** and external wireless interface **65**. Communication interface **62** provides communications with each of actuator control modules **18-21** via communication connection **37***f*. Communication interface **32** communicates data, commands and states. Processor **63** provides control and monitoring of modules **18-21**. Processor **63** receives commands and input via external communication interface **64** and feedback from modules **18-21**, via communication bus **60***a*, and provides command signals and controls modules **18-21**, via communication bus **60***a*, accordingly. Processor **61** is configured to perform a variety of computer-implemented functions such as performing method steps, calculations and the like and storing relevant data. Processor **61** may be any digital device which has output lines that are a logic function of its input lines, examples of which include a microprocessor, microcontroller, FPGA, PLD, application specific integrated circuit, or other similar devices.

(43) As shown, each of the four corners of housings **30***a*-**30***f* includes a longitudinally extending corner through-passage sized to receive four longitudinally extending tie rods **28***a*, **28***b*, **28***c* and **28***d*, respectively, therethrough. As shown. full assembly is completed aligning the four longitudinally extending corner through-passages in module housings **30***a*-**30***f* and using tie rods **28***a*, **28***b*, **28***c* and **28***d* therethrough to clamp module housings **30***a*-**30***f* together longitudinally with bolted end plates. Modules housings **30***a*-**30***f* seal against each other to provide a water and dust proof stacked assembly **15**. When assembled with rear covers **107***a*-**107***f* in place, rear compartments **99***a*-**99***f* form a common area that is open from module to module to allow

communication and power bus connections to be made internally to assembly **15**. In this embodiment, assembly **15** may have an Ingress Protection rating of at least IP67K. Tie rods **28***a*, **28***b*, **28***c* and **28***d* allow for the individual modules to be removable attached together in a compact stack and such that different modules may be swapped in and out and also such that modules may be added or removed from the stack as desired. While modular stack **15** is shown in this embodiment being held together with tie rod attachment **28***a*, **28***b*, **28***c* and **28***d*, other attachment systems may be employed as alternatives. For example, and without limitation, the modules may be individually connected to each other by bolts between adjacent modules, or other clamping or tensioning devices may be used to attach the modules in a coolant communicating stack.

(44) As shown, the left side of actuator module **18** includes two liquid coolant ports **80***a*, **82***a* and the right side of actuator module **18** includes two liquid coolant ports **81***a*, **83***a*. Horizontal interior coolant passage **85***a* extends directly between opposed ports **80***a* and **81***a*. In this embodiment, ports **80***a* and **81***a* are located in the upper left and right rear corners of module unit **18** and coolant passage **85***a* extends horizontally in the longitudinal direction x-x therebetween. Horizontal coolant passage **86***a* extends directly between opposed ports **82***a* and **83***a*. In this embodiment, ports **82***a* and **83***a* are located in the lower left and right front corners of module unit **18** and coolant passage **86***a* extends horizontally in longitudinal direction x-x therebetween. An interior power electronics coolant flow path **87***a* is provided between upper coolant passage **85***a* and lower coolant passage **86***a*. With reference to FIG. **6**, in this embodiment, the profile of coolant passage path **87***a* comprises rear vertical chamber **88** extending down from passage **85** in transverse direction y-y, front vertical chamber **89** extending up from passage **86** in transverse direction y-y, and a plurality of laterally extending and vertically spaced parallel coolant channels, severally indicated at **90**, that extend between rear vertical chamber **88***a* and front vertical chamber **89***a* and are located in close thermal proximity to power electronics **31***a*.

(45) As explained further below, each of ports **80***a*, **81***a*, **82***a* and **83***a* in housing **30***a* of module **18** are configured to be plugged to allow for alternate coolant configurations if desired. In addition, housing **30***a* of module **18** includes a number of alternate coolant ports to allow for the supply of coolant from alternate faces or sides of housing **30***a* of module **18** if desired. In particular, front port **91***a* is provided into interior coolant passage **86***a* from the front side **100***a* of module **18**, bottom port **92***a* is provided into interior coolant passage **86***a* from the bottom side **105***a* of module **18**, and top port **93***a* is provided into interior coolant passage **85***a* from the top side **104***a* of module **18**. Unless a particular alternate cooling configuration is desired, ports **91***a*, **92***a* and **93***a* are plugged, as shown in FIG. **1**.

(46) Similar to module **18**, the left side of actuator module **19** includes two liquid coolant ports **80***b*, **82***b* and the right side of actuator module **19** includes two liquid coolant ports **81***b*, **83***b*. Horizontal interior coolant passage **85***b* extends directly between opposed ports **80***b* and **81***b*. In this embodiment, ports **80***b* and **81***b* are located in the upper left and right rear corners of module unit **19** and coolant passage **85***b* extends horizontally in the longitudinal direction x-x therebetween. Horizontal coolant passage **86***b* extends directly between opposed ports **82***b* and **83***b*. In this embodiment, ports **82***b* and **83***b* are located in the lower left and right front corners of module unit **19** and coolant passage **86***b* extends horizontally in longitudinal direction x-x therebetween. An interior power electronics coolant flow path **87***b* is provided between upper coolant passage **85***b* and lower coolant passage **86***b*. In this embodiment, the profile of coolant passage path **87***b* is the same as coolant path **87***a* of module **18**, with rear vertical chamber **89** extending down from passage **85***b* in transverse direction y-y, front vertical chamber **89** extending up from passage **86***b* in transverse direction y-y, and a plurality of laterally extending and vertically spaced parallel coolant channels **90** extending between rear vertical chamber **88** and front vertical chamber **89** and located in close thermal proximity to power electronics **31***b*.

(47) As explained further below, each of ports **80***b*, **81***b*, **82***b* and **83***b* in housing **30***b* of module **19** are configured to be plugged to allow for alternate coolant configurations if desired. In additional,

housing **30***b* of module **19** includes a number of alternate coolant ports to allow for the supply of coolant from alternate faces or sides of housing **30***b* of module **19** if desired. In particular, front port **91***b* is provided into interior coolant passage **86***b* from the front side **100***b* of module **19**, bottom port **92***b* is provided into interior coolant passage **86***b* from the bottom side **105***b* of module **19**, and top port **93***b* is provided into interior coolant passage **85***b* from the top side **104***b* of module **19**. Unless a particular alternate cooling configuration is desired, ports **91***b*, **92***b* and **93***b* are plugged, as shown in FIG. **1**.

(48) Similar to module **19**, the left side of actuator module **20** includes two liquid coolant ports **80***c*, **82***c* and the right side of actuator module **20** includes two liquid coolant ports **81***c*, **83***c*. Horizontal interior coolant passage **85***c* extends directly between opposed ports **80***c* and **81***c*. In this embodiment, ports **80***c* and **81***c* are located in the upper left and right rear corners of module unit **20** and coolant passage **85***c* extends horizontally in the longitudinal direction x-x therebetween. Horizontal coolant passage **86***c* extends directly between opposed ports **82***c* and **83***c*. In this embodiment, ports **82***c* and **83***c* are located in the lower left and right front corners of module unit **20** and coolant passage **86***c* extends horizontally in longitudinal direction x-x therebetween. An interior power electronics coolant flow path **87***c* is provided between upper coolant passage **85***c* and lower coolant passage **86***c*. In this embodiment, the profile of coolant passage path **87***c* is the same as coolant path **87***a* of module **18**, with rear vertical chamber **89** extending down from passage **85***c* in transverse direction y-y, front vertical chamber **89** extending up from passage **86***c* in transverse direction y-y, and a plurality of laterally extending and vertically spaced parallel coolant channels **90** extending between rear vertical chamber **88** and front vertical chamber **89** and located in close thermal proximity to power electronics **31***c*.

(49) As explained further below, each of ports **80***c*, **81***c*, **82***c* and **83***c* in housing **30***c* of module **20** are configured to be plugged to allow for alternate coolant configurations if desired. In additional, housing **30***c* of module **20** includes a number of alternate coolant ports to allow for the supply of coolant from alternate faces or sides of housing **30***c* of module **20** if desired. In particular, front port **91***c* is provided into interior coolant passage **86***c* from the front side **100***c* of module **20**, bottom port **92***c* is provided into interior coolant passage **86***c* from the bottom side **105***c* of module **20**, and top port **93***c* is provided into interior coolant passage **85***c* from the top side **104***c* of module **20**. Unless a particular alternate cooling configuration is desired, ports **91***c*, **92***c* and **93***c* are plugged, as shown in FIG. **1**.

(50) Similar to module **20**, the left side of actuator module **21** includes two liquid coolant ports **80***d*, **82***d* and the right side of actuator module **21** includes two liquid coolant ports **81***d*, **83***d*. Horizontal interior coolant passage **85***d* extends directly between opposed ports **80***d* and **81***d*. In this embodiment, ports **80***d* and **81***d* are located in the upper left and right rear corners of module unit **21** and coolant passage **85***d* extends horizontally in the longitudinal direction x-x therebetween. Horizontal coolant passage **86***d* extends directly between opposed ports **82***d* and **83***d*. In this embodiment, ports **82***d* and **83***d* are located in the lower left and right front corners of module unit **21** and coolant passage **86***d* extends horizontally in longitudinal direction x-x therebetween. An interior power electronics coolant flow path **87***d* is provided between upper coolant passage **85***d* and lower coolant passage **86***d*. In this embodiment, the profile of coolant passage path **87***d* is the same as coolant path **87***a* of module **18**, with rear vertical chamber **89** extending down from passage **85***d* in transverse direction y-y, front vertical chamber **89** extending up from passage **86***d* in transverse direction y-y, and a plurality of laterally extending and vertically spaced parallel coolant channels **90** extending between rear vertical chamber **88** and front vertical chamber **89** and located in close thermal proximity to power electronics **31***e* and **31***f*.

(51) As explained further below, each of ports **80***d*, **81***d*, **82***d* and **83***d* in housing **30***d* of module **21** are configured to be plugged to allow for alternate coolant configurations if desired. In additional, housing **30***d* of module **21** includes a number of alternate coolant ports to allow for the supply of coolant from alternate faces or sides of housing **30***d* of module **21** if desired. In particular, front

port **91***d* is provided into interior coolant passage **86***d* from the front side **100***d* of module **21**, bottom port **92***d* is provided into interior coolant passage **86***d* from the bottom side **105***d* of module **21**, and top port **93***d* is provided into interior coolant passage **85***d* from the top side **104***d* of module **21**. Unless a particular alternate cooling configuration is desired, ports **91***d*, **92***d* and **93***d* are plugged, as shown in FIG. **1**.

(52) Similar to module **20**, the left side of central power control module **17** includes two liquid coolant ports **80***e*, **82***e* and the right side of central power control module **17** includes two liquid coolant ports **81***e*, **83***e*. Horizontal interior coolant passage **85***e* extends directly between opposed ports **80***e* and **81***e*. In this embodiment, ports **80***e* and **81***e* are located in the upper left and right rear corners of module unit **17** and coolant passage **85***e* extends horizontally in the longitudinal direction x-x therebetween. Horizontal coolant passage **86***e* extends directly between opposed ports **82***e* and **83***e*. In this embodiment, ports **82***e* and **83***e* are located in the lower left and right front corners of module unit **17** and coolant passage **86***e* extends horizontally in longitudinal direction x-x therebetween. An interior central power management electronics coolant flow path **87***e* is provided between upper coolant passage **85***e* and lower coolant passage **86***e*. In this embodiment, the profile of coolant passage path **87***e* is the same as coolant path **87***a* of module **18**, with rear vertical chamber **89** extending down from passage **85***e* in transverse direction y-y, front vertical chamber **89** extending up from passage **86***e* in transverse direction y-y, and a plurality of laterally extending and vertically spaced parallel coolant channels **90** extending between rear vertical chamber **88** and front vertical chamber **89** and located in close thermal proximity to central power management electronics **51**.

(53) As explained further below, each of ports **80***e*, **81***e*, **82***e* and **83***e* in housing **30***e* of module unit **17** are configured to be plugged to allow for alternate coolant configurations if desired. In additional, housing **30***e* of module unit **17** includes a number of alternate coolant ports to allow for the supply or discharge of coolant from alternate faces or sides of housing **30***e* of module **17** if desired. In particular, front port **91***e* is provided into interior coolant passage **86***e* from the front side **100***e* of module **17**, bottom port **92***e* is provided into interior coolant passage **86***e* from the bottom side **105***e* of module **17**, and top port **93***e* is provided into interior coolant passage **85***e* from the top side **104***e* of module **17**. Unless a particular alternate cooling configuration is desired, ports **92***e* and **93***e* are plugged. As explained further below, with central control module **16** stacked to the right of module **17** and in a certain combined series and parallel cooling configuration, alternate port **91***e* may be unplugged and employed as an outlet port, as shown in FIG. **1**.

(54) The porting and cooling passage design of modules **17-21** allows for the cooling flow paths of assembly **15** to be tailored to achieve the flow and pressure drop requirements for a particular application simply by the positioning of plugs. Thus, using the same modules, the flow paths can be series, parallel, or a combination of both series and parallel and may be tuned to specific requirements. FIG. **7** shows a series flow path cooling configuration, In the series cooling configuration shown in FIG. **7**, left upper side port **80***a* is plugged with plug **95***a* and right lower port **83***a* is blocked by plug **96***b*. Coolant is supplied via left lower port **82***a* and circulation pump **45** to passage **86***a* and, because of plug **96***b* blocking port **83***a*, into vertical chamber **89** of coolant flow path **87***a*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***a* and flows from the front to the rear of module **18** at the different heights of channels **90** to thereby cool power electronics **31***a*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the rear of channels **90** into rear vertical chamber **88** of coolant flow path **87***a*. Because of plug **95***a* in port **80***a*, the coolant is directed to exit module **18** via right upper port **81***a*. Accordingly, in this serial flow cooling configuration, module **18** receives coolant through a single inlet port **82***a* and discharges coolant through a single outlet port **81***a*.

(55) With modules **18** and **19** stacked side-to-side, right upper port **81***a* of module **18** is aligned and in sealed fluid communication with left upper port **80***b* of module **19**. Coolant thereby exits right

upper port **81***a* of module **18** into left upper port **80***b* of module **19**. Left lower side port **82***b* is plugged with plug **96***b* and right upper port **81***b* is blocked by plug **95***c*. Coolant is supplied via left upper port **80***b* and circulation pump **45** to passage **85***b* and, because of plug **95***c* blocking port **81***b*, into rear vertical chamber **88** of coolant flow path **87***b*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***b* and flows from the rear to the front of module **19** at the different heights of channels **90** to thereby cool power electronics **31***b*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***b*. Because of plug **96***b* in port **82***b*, the coolant is directed to exit module **19** via right lower port **83***b*. Accordingly, in this serial flow cooling profile, module **19** receives coolant through a single inlet port **80***b* and discharges coolant through a single outlet port **83***b*.

(56) With modules **19** and **20** stacked side-to-side, right lower port **83***b* of module **19** is aligned and in sealed fluid communication with left lower port **82***c* of module **20**. Coolant thereby exits right lower port **83***b* of module **19** into left lower port **82***c* of module **20**. Left upper side port **80***c* is plugged with plug **95***c* and right lower port **83***c* is blocked by plug **96***d*. Coolant is supplied via left lower port **82***c* and circulation pump **45** to passage **86***c* and, because of plug **96***d* blocking port **83***c*, into vertical front chamber **89** of coolant flow path **87***c*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***c* and flows from the front to the rear of module **20** at the different heights of channels **90** to thereby cool power electronics **31***c* and **31***d*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the rear of channels **90** into rear vertical chamber **88** of coolant flow path **87***b*. Because of plug **95***c* in port **80***c*, the coolant is directed to exit module **20** via right upper port **81***c*. Accordingly, in this serial flow cooling profile, module **20** receives coolant through a single inlet port **82***c* and discharges coolant through a single outlet port **81***c*.

(57) With modules **20** and **21** stacked side-to-side, right upper port **81***c* of module **20** is aligned and in sealed fluid communication with left upper port **80***d* of module **21**. Coolant thereby exits right upper port **81***c* of module **20** into left upper port **80***d* of module **21**. Left lower side port **82***d* is plugged with plug **96***d* and right upper port **81***d* is blocked by plug **95***e*. Coolant is supplied via left upper port **80***d* and circulation pump **45** to passage **85***d* and, because of plug **95***e* blocking port **81***d*, into rear vertical chamber **88** of coolant flow path **87***d*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***d* and flows from the rear to the front of module **19** at the different heights of channels **90** to thereby cool power electronics **31***f* and **31***g*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***d*. Because of plug **96***d* in port **82***d*, the coolant is directed to exit module **21** via right lower port **83***d*. Accordingly, in this serial flow cooling profile, module **21** receives coolant through a single inlet port **80***d* and discharges coolant through a single outlet port **83***d*.

(58) With modules **21** and **17** stacked side-to-side, right lower port **83***d* of module **21** is aligned and in sealed fluid communication with left lower port **82***e* of module **17**. Coolant thereby exits right lower port **83***d* of module **21** into left lower port **82***e* of module **17**. Left upper side port **80***e* is plugged with plug **95***e* and right lower port **83***e* is blocked by plug **96***f*. Coolant is supplied via left lower port **82***e* and circulation pump **45** to passage **86***e* and, because of plug **96***f* blocking port **83***e*, into vertical front chamber **89** of coolant flow path **87***e*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***e* and flows from the front to the rear of module **17** at the different heights of channels **90** to thereby cool central power management electronics **51**, including control power electronics **55**, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the rear of channels **90** into rear vertical chamber **88** of coolant flow path **87***c*. Because of plug **95***e* in port **80***e*, the coolant is directed to exit module **17** via right upper port **81***e*. Accordingly, in this serial flow cooling profile, module **20** receives coolant through a single inlet port **82***e* and discharges coolant through a single

outlet port **81***e*. In the embodiment shown in FIG. **1**, with central control module **16** stacked to the right of module **17** and because central control module **16** does not include any coolant channels in this embodiment, port **81***e* may be plugged and alternate port **93***e* may be unplugged and employed as an outlet port. Accordingly, in this serial flow cooling profile, module **17** receives coolant through a single inlet port **82***e* and discharges coolant through a single outlet port **81***e* or **93***e*.
(59) FIG. **8** shows a parallel flow path cooling configuration. In the parallel cooling configuration shown in FIG. **8**, left lower side port **82***a* is plugged with plug **96***a*. Neither of right upper port **81***a* or right lower port **83***a* of module **18** are plugged. Also, neither of left upper port **80***b* or left lower port **82***b* of module **19** are plugged. Coolant is supplied via left upper port **80***a* and circulation pump **45** to passage **85***a*. Coolant is directed to flow both through passage **85***a* to right upper port **81***a* as well as into vertical rear chamber **88** of coolant flow path **87***a*. From vertical rear chamber **88** of coolant flow path **87***a*, coolant flows into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***a* and flows from the rear to the front of module **18** at the different heights of channels **90** to thereby cool power electronics **31***a*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***a*. Because of plug **96***a* in port **82***a*, the coolant is directed to exit module **18** via either right upper port **81***a* or right lower port **83***a*. Accordingly, in this parallel flow cooling configuration, module **18** receives coolant through at least inlet port **80***a* and discharges coolant through two parallel outlet ports **81***a* and **83***a*.
(60) With modules **18** and **19** stacked side-to-side, right upper port **81***a* of module **18** is aligned and in sealed fluid communication with left upper port **80***b* of module **19**. Coolant thereby exits right upper port **81***a* of module **18** into left upper port **80***b* of module **19**. Also, right lower port **83***a* of module **18** is aligned and in sealed fluid communication with left lower port **82***b* of module **19**. Coolant thereby also exits right lower port **83***a* of module **18** into left lower port **82***b* of module **19**. Neither of right upper port **81***b* or right lower port **83***b* of module **19** are plugged. Also, neither of left upper port **80***c* or left lower port **82***c* of module **20** are plugged. Coolant is supplied via left upper port **80***b* to passage **85***b* and via left lower port **82***b* to passage **86***b*. Coolant is directed to flow through passage **85***b* to right upper port **81***b*, through passage **86***b* to right lower port **83***b*, as well as into channels **90** of coolant flow path **87***b* to thereby cool power electronics **31***b*. Accordingly, in this parallel flow cooling configuration, module **19** receives coolant through two parallel inlet ports **80***b* and **82***b* and discharges coolant through two parallel outlet ports **81***b* and **83***b*.
(61) With modules **19** and **20** stacked side-to-side, right upper port **81***b* of module **19** is aligned and in sealed fluid communication with left upper port **80***c* of module **20**. Coolant thereby exits right upper port **81***b* of module **19** into left upper port **80***c* of module **20**. Also, right lower port **83***b* of module **19** is aligned and in sealed fluid communication with left lower port **82***c* of module **20**. Coolant thereby also exits right lower port **83***b* of module **19** into left lower port **82***c* of module **20**. Neither of right upper port **81***c* or right lower port **83***c* of module **20** are plugged. Also, neither of left upper port **80***c* or left lower port **82***d* of module **21** are plugged. Coolant is supplied via left upper port **80***c* to passage **85***c* and via left lower port **82***c* to passage **86***c*. Coolant is directed to flow through passage **85***c* to right upper port **81***c*, through passage **86***c* to right lower port **83***c*, as well as into channels **90** of coolant flow path **87***c* to thereby cool power electronics **31***c* and **31***d*. Accordingly, in this parallel flow cooling configuration, module **20** receives coolant through two parallel inlet ports **80***c* and **82***c* and discharges coolant through two parallel outlet ports **81***c* and **83***c*.
(62) With modules **20** and **21** stacked side-to-side, right upper port **81***c* of module **20** is aligned and in sealed fluid communication with left upper port **80***d* of module **21**. Coolant thereby exits right upper port **81***c* of module **20** into left upper port **80***d* of module **21**. Also, right lower port **83***c* of module **20** is aligned and in sealed fluid communication with left lower port **82***d* of module **21**. Coolant thereby also exits right lower port **83***c* of module **20** into left lower port **82***d* of module **21**.

Neither of right upper port **81***d* or right lower port **83***d* of module **21** are plugged. Also, neither of left upper port **80***e* or left lower port **82***e* of module **17** are plugged. Coolant is supplied via left upper port **80***d* to passage **85***d* and via left lower port **82***d* to passage **86***d*. Coolant is directed to flow through passage **85***d* to right upper port **81***d*, through passage **86***d* to right lower port **83***d*, as well as into channels **90** of coolant flow path **87***d* to thereby cool power electronics **31***e* and **31***f*. Accordingly, in this parallel flow cooling configuration, module **21** receives coolant through two parallel inlet ports **80***d* and **82***d* and discharges coolant through two parallel outlet ports **81***d* and **83***d*.

(63) With modules **21** and **17** stacked side-to-side, right upper port **81***d* of module **21** is aligned and in sealed fluid communication with left upper port **80***e* of module **17**. Coolant thereby exits right upper port **81***d* of module **21** into left upper port **80***e* of module **17**. Also, right lower port **83***d* of module **21** is aligned and in sealed fluid communication with left lower port **82***e* of module **17**. Coolant thereby also exits right lower port **83***d* of module **21** into left lower port **82***e* of module **17**. Coolant is supplied via left upper port **80***e* to passage **85***e* and via left lower port **82***e* to passage **86***e*. Right upper side port **81***e* is plugged with plug **95***f*. Coolant is directed to flow both through passage **86***e* to right lower port **83***e* as well as through passage **85***e* into vertical rear chamber **88** of coolant flow path **87***e*. From vertical rear chamber **88** of coolant flow path **87***e*, coolant flows into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***e* and flows from the rear to the front of module **17** at the different heights of channels **90** to thereby cool electronics **51**, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***e* and exits module **17** via right lower port **83***e*. In the embodiment shown in FIG. **1**, with central control module **16** stacked to the right of module **17** and because central control module **16** does not include any coolant channels in this embodiment, port **83***e* may be plugged and alternate port **91***e* may be unplugged and employed as an outlet port. Accordingly, in this parallel flow cooling configuration, module **17** receives coolant through two parallel inlet ports **80***e* and **82***e* and discharges coolant through at least outlet port **83***e* or **91***e*.

(64) FIG. **9** shows a combined series and parallel flow path cooling configuration, In the combined series and parallel cooling configuration shown in FIG. **9**, left upper side port **80***a* is plugged with plug **95***a* and right lower port **83***a* is blocked by plug **96***b*. Coolant is supplied via left lower port **82***a* and circulation pump **45** to passage **86***a* and, because of plug **96***b* blocking port **83***a*, into vertical chamber **89** of coolant flow path **87***a*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***a* and flows from the front to the rear of module **18** at the different heights of channels **90** to thereby cool power electronics **31***a*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the rear of channels **90** into rear vertical chamber **88** of coolant flow path **87***a*. Because of plug **95***a* in port **80***a*, the coolant is directed to exit module **18** via right upper port **81***a*. Accordingly, module **18** is in a serial flow cooling configuration and receives coolant through a single inlet port **82***a* and discharges coolant through a single outlet port **81***a*.

(65) With modules **18** and **19** stacked side-to-side, right upper port **81***a* of module **18** is aligned and in sealed fluid communication with left upper port **80***b* of module **19**. Coolant thereby exits right upper port **81***a* of module **18** into left upper port **80***b* of module **19**. Left lower side port **82***b* is plugged with plug **96***b* and right upper port **81***b* is blocked by plug **95***c*. Coolant is supplied via left upper port **80***b* and circulation pump **45** to passage **85***b* and, because of plug **95***c* blocking port **81***b*, into rear vertical chamber **88** of coolant flow path **87***b*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***b* and flows from the rear to the front of module **19** at the different heights of channels **90** to thereby cool power electronics **31***b*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***b*. Because of plug **96***b* in port **82***b*, the coolant is directed to exit module **19** via right lower port **83***b*. Accordingly,

modules **18** and **19** are in a serial flow cooling configuration and module **19** receives coolant through a single inlet port **80***b* and discharges coolant through a single outlet port **83***b*.

(66) With modules **19** and **20** stacked side-to-side, right lower port **83***b* of module **19** is aligned and in sealed fluid communication with left lower port **82***c* of module **20**. Coolant thereby exits right lower port **83***b* of module **19** into left lower port **82***c* of module **20**. Left upper side port **80***c* is plugged with plug **95***c* and right lower port **83***c* is blocked by plug **96***d*. Coolant is supplied via left lower port **82***c* and circulation pump **45** to passage **86***c* and, because of plug **96***d* blocking port **83***c*, into vertical front chamber **89** of coolant flow path **87***c*. Coolant is then directed into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***c* and flows from the front to the rear of module **20** at the different heights of channels **90** to thereby cool power electronics **31***c* and **31***d*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the rear of channels **90** into rear vertical chamber **88** of coolant flow path **87***b*. Because of plug **95***c* in port **80***c*, the coolant is directed to exit module **20** via right upper port **81***c*. Accordingly, modules **18**, **19** and **20** are in a serial flow cooling configuration and module **20** receives coolant through a single inlet port **82***c* and discharges coolant through a single outlet port **81***c*.

(67) With modules **20** and **21** stacked side-to-side, right upper port **81***c* of module **20** is aligned and in sealed fluid communication with left upper port **80***d* of module **21**. Coolant thereby exits right upper port **81***c* of module **20** into left upper port **80***d* of module **21**. Left lower side port **82***d* is plugged with plug **96***d*. Neither of right upper port **81***d* or right lower port **83***d* of module **21** are plugged. Also, neither of left upper port **80***e* or left lower port **82***e* of module **17** are plugged. Coolant is supplied via left upper port **80***d* and circulation pump **45** to passage **85***d*. Coolant is directed to flow both through passage **85***d* to right upper port **81***d* as well as into vertical rear chamber **88** of coolant flow path **87***d*. From vertical rear chamber **88** of coolant flow path **87***d*, coolant flows into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***d* and flows from the rear to the front of module **21** at the different heights of channels **90** to thereby cool power electronics **31***e* and **31***f*, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***d*. Because of plug **96***d* in port **82***d*, the coolant is directed to exit module **21** via either right upper port **81***d* or right lower port **83***d*. Accordingly, module **18** is in a parallel flow cooling configuration and receives coolant through at least inlet port **80***d* and discharges coolant through two parallel outlet ports **81***d* and **83***d*.

(68) With modules **21** and **17** stacked side-to-side, right upper port **81***d* of module **21** is aligned and in sealed fluid communication with left upper port **80***e* of module **17**. Coolant thereby exits right upper port **81***d* of module **21** into left upper port **80***e* of module **17**. Also, right lower port **83***d* of module **21** is aligned and in sealed fluid communication with left lower port **82***e* of module **17**. Coolant thereby also exits right lower port **83***d* of module **21** into left lower port **82***e* of module **17**. Coolant is supplied via left upper port **80***e* to passage **85***e* and via left lower port **82***e* to passage **86***e*. Right upper side port **81***e* is plugged with plug **95***f*. Coolant is directed to flow both through passage **86***e* to right lower port **83***e* as well as through passage **85***e* into vertical rear chamber **88** of coolant flow path **87***e*. From vertical rear chamber **88** of coolant flow path **87***e*, coolant flows into each of horizontal laterally extending parallel channels **90** of coolant flow path **87***e* and flows from the rear to the front of module **17** at the different heights of channels **90** to thereby cool electronics **51**, which are located in close thermal proximity to cooling channels **90**. The fluid coolant exits the front of channels **90** into front vertical chamber **89** of coolant flow path **87***e* and exits module **17** via right lower port **83***e*. In the embodiment shown in FIG. **1**, with central control module **16** stacked to the right of module **17** and because central control module **16** does not include any coolant channels in this embodiment, port **83***e* may be plugged and alternate port **91***e* may be unplugged and employed as an outlet port. Accordingly, module **21** and **17** are in a parallel flow cooling configuration and module **17** receives coolant through two parallel inlet ports **80***e* and **82***e*

and discharges coolant through at least outlet port **83***e* or **91***e*.
(69) FIG. **11** shows a first alternative cooling path **187** to the cooling path **87** shown in FIG. **6**. In this embodiment, the left side of the module also includes upper liquid coolant port **180** and lower liquid coolant port **182**. The right side of the module includes a first liquid coolant port aligned opposite to coolant port **180** with coolant passage **185** extending horizontally in the longitudinal direction x-x therebetween, and the right side of the module includes a second liquid coolant port aligned opposite to coolant port **182** with coolant passage **186** extending horizontally in the longitudinal direction x-x therebetween. Cooling passage **185** is located in the upper left corner of module unit **118**. However, in this embodiment coolant passage **186** is located in the lower left corner of the module unit. As shown, in this embodiment interior power electronics coolant flow path **187** extends as a single serpentine path between upper coolant passage **185** and lower coolant passage **186***a*.
(70) FIG. **12** shows a second alternative cooling path **287** to the cooling path **87** shown in FIG. **6**. As with the embodiment shown in FIG. **11**, in this embodiment, the left side of the module includes upper liquid coolant port **280** and lower liquid coolant port **282**. The right side of the module includes a first liquid coolant port aligned opposite to coolant port **280** with coolant passage **285** extending horizontally in the longitudinal direction x-x therebetween, and the right side of the module includes a second liquid coolant port aligned opposite to coolant port **282** with coolant passage **286** extending horizontally in the longitudinal direction x-x therebetween. Cooling passage **285** is located in the upper left corner of the module unit and coolant passage **286** is located in the lower left corner of module unit **118**. As shown, in this embodiment the profile of coolant passage path **287** comprises coolant channel **291** extending laterally in direction z-z from passage **285**, front vertical chamber **289** extending down from passage **291** in transverse direction y-y, rear vertical chamber **288** extending up from passage **286** in transverse direction y-y, and a plurality of laterally extending and vertically spaced parallel coolant channels, severally indicated at **290**, that extend between front vertical chamber **289** and rear vertical chamber **188**.
(71) While the coolant paths are shown with various geometries and conduit configurations, alternative passage geometries and porting may be used. For example, a partitioned volume of the interior of the housing may be used to provide the coolant passage or the input and output ports may be located in alternative locations through the housing and the number of such ports may be varied as desired.
(72) Modular control system **15** has a number of advantages. System **15** provides a stack of individualized electronics modules that is liquid cooled, that is very compact, that is environmentally rugged, that is mechanically robust, that is expandable, and that is rated for environments such as compact earth moving equipment. System **15** is easily scalable and customizable. The integrated cooling passages designed into each module eliminate the need for external interconnections between them. Integrated electric bus connections, both power and control, eliminating the need for external interconnections between them. The individual stacked modular units may be customized to provide individualized desired control electronics. The number and configuration of the modular units may be varied as desired for the application and conditions of the environment. The system provides IoT data gathering, storage and transmission. The system has remote control capability and has autonomous control capability. The individual module units may also be line replaceable units (LRUs). The individual modules may have an Ingress Protection rating of at least IP44 and when placed into an assembly of multiple modules may have an Ingress Protection rating of at least IP67K. System **15** is scalable in size by adding module units to the stack as needed.
(73) While the presently preferred form of the modular actuator control system has been shown and described, and several modifications thereof discussed, persons skilled in this art will readily appreciate that various additional changes and modifications may be made without departing from the scope of the invention, as defined and differentiated by the claims.

Claims

1. A modular actuator control system, comprising: a first controller module configured to control an electrically powered first actuator having at least one motion axis; a second controller module configured to control an electrically powered second actuator having at least one motion axis; said first controller module comprising: a first housing; first control electronics in said first housing; first power electronics in said first housing; a first power connection configured to connect to a master power source; a first communication connection configured to connect to master controller electronics; a first actuator power connection configured to connect said first power electronics with said first actuator; a first coolant inlet port; a first coolant outlet port; and a first flow path in said first housing between said first coolant inlet port and said first coolant outlet port that is configured to provide a liquid coolant to said first power electronics; said second controller module comprising: a second housing; second control electronics in said second housing; second power electronics in said second housing; a second power connection configured to connect to said power source; a second communication connection configured to connect to said master controller electronics; a second actuator power connection configured to connect said second power electronics with said second actuator; a second coolant inlet port; a second coolant outlet port; and a second flow path in said second housing between said second coolant inlet port and said second coolant outlet port that is configured to provide said liquid coolant to said second power electronics; an attachment connecting said first housing of said first controller module and said second housing of said second controller module; said first coolant inlet port of said first controller module configured to connect to a fluid coolant source; and said first coolant outlet port of said first controller module connected to said second coolant inlet port of said second controller module; wherein said first controller module and said second controller module are stacked in coolant fluid communication with each other.
2. The modular actuator control system set forth in claim 1, wherein: said first controller module comprises a first secondary coolant outlet port and a first secondary flow path in said first housing between said first coolant inlet port and said first secondary coolant outlet port; said first flow path is separate from said first secondary flow path; said second controller module comprises a second secondary coolant inlet port; and said first secondary coolant outlet port of said first controller module is connected to said second secondary coolant inlet port of said second controller module; wherein said first controller module and said second controller module are stacked in a parallel flow path coolant configuration.
3. The modular actuator control system set forth in claim 1, wherein: said first controller module comprises a first actuator communication connection configured to connect said first controls electronics with said first actuator; and said second controller module comprises a second actuator communication connection configured to connect said second controls electronics with said second actuator.
4. The modular actuator control system set forth in claim 3, wherein: said first actuator comprises a first sensor for sensing an operating parameter of said first actuator and said first actuator communication connection is configured to connect said first controls electronics with said first sensor of said first actuator; and said second actuator comprises a second sensor for sensing an operating parameter of said second actuator and said second actuator communication connection is configured to connect said second controls electronics with said second sensor of said second actuator.
5. The modular actuator control system set forth in claim 1, comprising a common power bus supplying electric power to said first power connection and said second power connection.
6. The modular actuator control system set forth in claim 1, comprising a common serial bus communicating with said first communication connection and said second communication

connection.
7. The modular actuator control system set forth in claim 1, wherein: said first housing comprises a first sealed electronics housing section defining a first electronics chamber substantially isolated from an outside environment and said first control electronics and said first power electronics are disposed in said first chamber; and said second housing comprises a second sealed electronics housing section defining a second electronics chamber substantially isolated from said outside environment and said second control electronics and said second power electronics are disposed in said second chamber.
8. The modular actuator control system set forth in claim 7, wherein: said first housing comprises a first connection section defining a first connection chamber substantially isolated from said outside environment and said first power connection and said first communication connection are disposed in said first connection chamber; and said second housing comprises a second connection section defining a second connection chamber substantially isolated from said outside environment and said second power connection and said second communication connection are disposed in said second connection chamber.
9. The modular actuator control system set forth in claim 8, comprising: a power bus supplying electric power to said first power connection and said second power connection; a serial bus communicating with said first communication connection and said second communication connection; and said common power bus extending into said first connection chamber and said second connection chamber; and said common serial bus extending into said first connection chamber and said second connection chamber.
10. The modular actuator control system set forth in claim 1, comprising: a master controller module comprising: a master controller housing; said master controller electronics disposed in said master controller housing; a master communication connection configured to connect to said first communication connection of said first controller module and said second communication connection of said second controller module; and said attachment connecting said first housing of said first controller module, said second housing of said second controller module, and said master controller housing of said master controller module.
11. The modular actuator control system set forth in claim 10, comprising a serial bus communicating between said master communication connection, said first communication connection and said second communication connection.
12. The modular actuator control system set forth in claim 10, comprising: a power management module comprising: a power management housing; power management electronics in said power management housing; an input power connection configured to connect to said master power source; an output control power connection configured to connect to said first control electronics and said second control electronics; a power coolant inlet port; a power coolant outlet port; and a power flow path in said power management housing between said power coolant inlet port and said power coolant outlet port that is configured to provide a liquid coolant to said power management electronics; and said attachment connecting said first housing of said first controller module, said second housing of said second controller module, said master controller housing of said master controller module, and said power management housing of said power management module.
13. The modular actuator control system set forth in claim 12, wherein said master controller module comprises an input control power connection configured to connect to said output control power connection of said power management module.
14. The modular actuator control system set forth in claim 1, comprising a pump connected to said first coolant inlet port of said first controller module and operatively configured to pump said liquid coolant through said first flow path of said first housing and said second flow path of said second housing.
15. The modular actuator control system set forth in claim 14, comprising a heat exchanger connected between said second coolant outlet port of said second controller module and said first

coolant inlet port of said first controller module.
16. The modular actuator control system set forth in claim 15, wherein said heat exchanger is connected between said second coolant outlet port of said second controller module and said pump.
17. The modular actuator control system set forth in claim 1, comprising a seal between said first housing of said first controller module and said second housing of said second controller module.
18. The modular actuator control system set forth in claim 12, wherein said power management module is stacked between said second controller module and said master controller module and wherein said second coolant outlet port of said second controller module is connected to said power coolant inlet port of said power management module.
19. The modular actuator control system set forth in claim 1, comprising: a third controller module configured to control an electrically power third actuator having at least one motion axis; said third controller module comprising: a third housing; third control electronics in said third housing; third power electronics in said third housing; a third power connection configured to connect to said power source; a third communication connection configured to connect to said master controller electronics; a third actuator power connection configured to connect said third power electronics with said third actuator; a third coolant inlet port; a third coolant outlet port; and a third flow path in said third housing between said third coolant inlet port and said third coolant outlet port that is configured to provide said liquid coolant to said third power electronics; said attachment connecting said first housing of said first controller module, said second housing of said second controller module, and said third housing of said third controller module; and said second coolant outlet port of said second controller module connected to said third coolant inlet port of said third controller module; wherein said first controller module, said second controller module, and said third controller module are stacked in coolant fluid communication with each other.
20. The modular actuator control system set forth in claim 19, wherein: said second controller module comprises a second secondary coolant outlet port and a second secondary flow path in said second housing between said second coolant inlet port and said second secondary coolant outlet port; said second flow path is separate from said second secondary flow path; said third controller module comprises a third secondary coolant inlet port; and said second secondary coolant outlet port of said second controller module is connected to said third secondary coolant inlet port of said third controller module; wherein said first controller module and said second controller module are stacked in a series flow path coolant configuration and said second controller module and said third controller module are stacked in a parallel flow path coolant configuration.
21. The modular actuator control system set forth in claim 1, wherein said first flow path in said first housing between said first coolant inlet port and said first coolant outlet port comprises a plurality of separate coolant passages in thermal proximity to said first power electronics.
22. The modular actuator control system set forth in claim 1, wherein said first flow path in said first housing between said first coolant inlet port and said first coolant outlet port comprises a serpentine coolant passage in thermal proximity to said first power electronics.
23. The modular actuator control system set forth in claim 12, wherein said master controller housing of said master controller module contains a braking resistor and a capacitor.
24. The modular actuator control system set forth in claim 12, wherein said power source comprises an electric-vehicle battery.